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Leveraging Carbon Markets for Clean Energy Transition: Evaluating the UAE National Carbon Credit Registry (NRCC) as a Catalyst for Renewable Energy Projects Deployment

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Abstract

This paper assesses the National Carbon Credit Registry (NRCC) created in the United Arab Emirates in 2024, evaluating its performance in bolstering the trade and funding of renewable energy initiatives. A techno-economic model of a 100 MW photovoltaic project is used in the analysis of carbon revenues at $10-200 per ton of CO_2, and a sensitivity study on discount rates (6%, 8%, 10%), the analysis clearly shows that increasing of carbon revenues (100-200$) will substantially increase project profitability especially when discount rate is low (6, 8%) as compared to rise in discount rates (8-10%). The case study results show that by using carbon pricing, LCOE could be reduced by 30%, IRR could increase 8-12%, and the discounted payback is remarkably decreased. Comparing the European Union Emissions Trading System (EU ETS) reveals marked differences in market maturity, regulation, and price volatility risk, primarily because a fully mature trading mechanism is challenging to establish within an emerging market. Though the NRCC works to strengthen the bankability of the renewable energy investment, the paper identifies some policy refinements that are vital in enhancing the stability of markets, as well as attracting longer-term private investment and guaranteeing long-term sustainability of the UAE carbon trading system development, including a standardization of baselines and placing price floors on carbon.

Introduction

UAE is now one of the most progressive nations in the Arab world in addressing climate change since it is now adopting a strategy towards Net Zero Emissions (NZE) 2050 under the First Sustainable Development Plan 2071, which is aligned with the Paris Agreement [5]. Besides adopting energy programs, the UAE has also developed strategic goals towards attaining sustainable energy in the future. These are meant to cut down GHG pollution and promote sustainable economic growth. Nevertheless, reducing emissions and achieving deep decarbonization of all sectors need not only an emphasis on technological innovations and flows of investments but also the use of market instruments that play a critical role in planning and mitigation.

Carbon markets and carbon credit registries have proved to be critical in allowing countries to cut emissions more cheaply. These systems are accompanied by costs of emission reduction, further emphasizing that they play the role of being environmentally responsible. In this regard, 2024 shall be a landmark year in terms of the UAE's carbon credit policies, with the creation of the National Carbon Credit Registry (NRCC). By enhancing the capacity of this country to achieve its climate targets and also making the NRCC a demonstration of a verifiable and scalable domestic carbon trading system, this initiative is helping to raise the country's capacity to achieve its climate targets, as well as giving the NRCC a paradigm of how verifiable and scalable domestic carbon trading should be.

The NRCC was gazetted by the UAE Cabinet Resolution No. 67 of 2024. The distribution, registration, and trading of carbon credits within the UAE are under the NRCC. It is a critical factor in growing the global market for voluntary carbon markets, which can foster international cooperation in carbon trading. Together with its efforts in reducing emissions, the NRCC promotes investments in renewable energy resources, including the construction of power plants based on renewable energy sources, to further facilitate the decrease in its carbon emissions [11]. This is the highlight of the significant presence of the NRCC as the meeting point between climate policy, sustainable finance, and the energy transition. As a mediator of carbon credit creation and trading, the NRCC plays a central role in the UAE's Net Zero agenda pursuit and contributes materially to the global carbon credit ecosystem [5].

This research paper explores the NRCC's contribution to promoting renewable energy projects in the UAE and examines alternative solutions that could further advance the implementation of effective renewable energy development.

Literature Review

Introduction of carbon markets and the inclusion of carbon credit systems in national energy policies have evoked much concern, especially among academics, owing to the capital required to undertake renewable energy schemes. The importance of carbon markets in facilitating the development of low-carbon infrastructure has been discussed in the context of existing literature, particularly in developing countries. As an example, Smith et al. (2020) presented a case study showing that the Levelized Cost of Electricity (LCOE) of renewable energy can be reduced by up to 30 % by including carbon credits in the financial model of these projects, thereby making them more appealing. The finding is especially applicable to countries such as the UAE, where significant capital expenditure (CAPEX) funds are at disposal [19]. Notwithstanding, initial costs can pose a bottleneck to large-scale projects in renewable energy. Carbon credits can be critical in helping phase out this obstacle by making such projects more economically sound, therefore making them more attractive to invest in.

In addition, UNFCCC CDM (2024) evaluated the relevance of carbon credit systems in different situations, such as the case of Europe and emerging markets. The carbon markets that often already exist (as per the example of the European Union) have proved that the cost of capital for solar energy projects has cratered. They have determined that with the carbon pricing mechanism, the CAPEX could be reduced by up to 15%, which seems to be a possibility of transitioning to renewable energy [5]. These systems are soon becoming proactive in the line of voluntary and compliance-oriented solutions to carbon trade, mainly due to the fact that the UAE is trying to transition to a sustainable and low-carbon economy.

The introduction of the NRCC in 2024 will be an outstanding achievement in the UAE since it will be the first carbon market developed nationally in the country. This project is central to assistance toward the 2050 Net Zero Emissions (NZE) strategy in the UAE. In time, the capacity of the NRCC to enhance the price of project financing and maintain the possibility of investing in renewable energy in the future is vital, as it will allow the generation and sale of carbon offsets. This domestic initiative supports the international effort to develop a transparent and efficient carbon market, facilitating investment in clean energy.

The literature on carbon credit systems is relatively developed, and much of this is in developed economies in which markets have already been established and nurtured. Nonetheless, research is scarce on how these systems can be modified to suit emerging economies, such as the UAE. The most promising areas for expansion include carbon price volatility, fluctuations in MRV (Monitoring, Reporting, and Verification) costs, and the efficiency of voluntary carbon markets in developing economies. Additionally, there is little debate about why voluntary carbon markets and compliance systems are effective in such countries. More studies should be conducted to address these obstacles without jeopardizing the compatibility of domestic markets with international standards [17].

Policy Framework

- **UAE Carbon Market Structure: Voluntary vs. Compliance Markets**

At this point, the carbon market is mainly based on companies voluntarily participating in the Duncan Seattle, UAE, and treating it as an opportunity to offset their emissions in an attempt to achieve sustainability goals or improve their ESG scores. Renewable energy, energy efficiency, and nature-based projects can produce carbon credits, provided they adhere to approved methodologies for verification. It is worth noting that these hubs have yet to be remedied to enforce compulsory registration under NRCC by these agencies, which continue to have significant carbon emissions unless they engage in voluntary participation in the carbon credit market.

Even though currently there is no compliance carbon market in the UAE, the introduction of NRCC is the first step in creating one. The registry aims to establish a two-way relationship with Emissions Trading Systems (ETS) or carbon taxes, thereby supporting the UAE's Net-Zero by 2050 targets. The NRCC has a twofold concentration on the development of market solutions and the generation of competitiveness to guarantee compliance with the market in the long term. The strategy indicates the proactive interests of the UAE in carbon governance, with the NRCC being at the center of the dynamic process in the three-way relationship between climate pledges, investments, and renewable energy investments [19].

In the existing model, any company can participate in the voluntary carbon market to balance its emissions, achieve its sustainability objectives, or enhance its ESG rating. It is, however, mandatory by law that entities with high carbon emissions must register at the NRCC. To stay within regulatory guidelines, these entities are expected to provide detailed reports of their activities, operations, and measures to reduce carbon emissions, and publish a statement of updated emissions.

Voluntary Carbon Market (Current System)

Table 1—Voluntary Carbon Market

Aspect	Description
Definition	A market mechanism where entities voluntarily purchase bon credits to offset emissions without being subject to ulatory obligations
Primary Purpose	To fulfil corporate sustainability objectives, meet ESG mmitments, and improve brand reputation.
Real-World Application	A UAE-based company may invest in a solar farm or ngrove restoration project to generate carbon credits for setting operational emissions.
Governance and frastructure	Managed by the (NRCC), which ensures verification, nsparent issuance, and prevents double-counting
Market Participants	Corporations, tourism operators, airlines, and government- ked entities seeking voluntary offsets
Regulatory Oversight	Supervised by MOCCAE under a light-touch regulatory proach that allows the market to drive participation

Compliance Carbon Market (Future Possibility)

Table 2—Compliance Carbon Market

Aspect	Description
Definition	A legally binding system where companies must reduce issions or purchase credits to comply with government- posed emission limits
Primary Purpose	To enforce national climate targets under the UAE's t Zero by 2050 plan
Expected Features	High-emitting sectors (e.g., oil & gas, power neration, cement) may be required to report and offset issions. The government may issue emissions caps and nalties for non-compliance
Registry Role	The NRCC may evolve to support a compliance market integrating allowance tracking and legal enforcement pabilities.
Market Participants	Regulated industries with mandatory reporting and mpliance obligations
Regulatory Oversight	High—managed through federal law or emissions ding legislation

Methodology

The potential of NCRR of the UAE to receive the investments in the business as well as the rise of the realization of renewable energy is to be determined by means of this complex assessment. The strengths and weaknesses of the NRCC are discussed here using policy analysis, international benchmarking and economical cost analysis to help in identifying the level in which the NRCC affects the projects of renewable energy.

The political and literary background of how the NRCC is to be constituted and administered has first been given. This is, in part, a comparison to established documents, including the Net Zero 2050 Strategy of the UAE [1], Article 6 of the Paris Agreement [5], and the processes of popular carbon credit systems such as the Verified Carbon Standard (VCS) [3] and the Gold Standard [4]. As it is designed according to the NRCC, the authors have checked whether these systems coincided with the climate and energy performance objectives of the UAE. Namely, the role of the NRCC in carbon credit trading was addressed with references to the regulations it offered and its compatibility with international public regulations. It further discusses the challenge of how the NRCC will assist the UAE in switching to clean energy by enabling the carbon credit market.

Second, the study used the financial case study of a 100 MW solar (PV) project in the UAE and assessed the feasibility of the renewable energy projects, which is based on the alternative carbon pricing scenario. The authors took note of how the project revenues were achieved, considering the carbon credits. Would impact the business model of the project and its impact on (IRR), recovery time, and (LCOE). The different carbon prices applied in the financial modelling ranged between 10 and 120 per ton of CO_2. In addition, the sensitivity analysis was carried out by considering the discount rate as 6%, 10% and 10% with the notion that this would show how the carbon pricing would be used in determining the profitability of the projects and the Return on Investment [12].

Third, it embarked on a comparative analysis as a way of evaluating the carbon credit regime of the remaining nations, particularly those of the EU. We were in a position to learn a lot more about the differences between the carbon credit markets in the EU, which is in a well-developed state, and the carbon trade in the developing state of the UAE. In their studies, the authors consider the impacts of these kinds of systems on the financial implications of the renewable energy projects and the prospect of scaling carbon credit instruments in the UAE. The best global practices were incorporated into the study to gauge the strategy of NRCC and determine the context under which the carbon credit market in the UAE could have a better strategy.

Studies show that the security of renewable energy projects can be enhanced by NRCC tremendously. By facilitating the market for carbon credits produced by renewable energy projects, the NRCC will enhance project bankability and promote the energy transition in the UAE. Nevertheless, the NRCC remains underutilized due to high verification costs, market fluctuations, and outstanding regulatory issues. More policies will therefore need to be implemented to stabilize the market and create an experience of more predictable carbon revenues.

Integration with Renewable Energy Projects under the NRCC

The (NRCC) plays a significant role in identifying renewable energy projects in the UAE and enabling them to earn more revenues by selling the carbon credits they create. These carbon offset projects need to meet the international standard of carbon offset. This is because they must be able to show real emission reductions that are credible in the environment.

i. **Credit Generation Process for Renewable Projects**

Renewable energy sources such as solar plants, windmills, and biogas plants in the UAE are eligible to generate carbon credits if they offset conventional sources of energy. However, to be eligible, these projects must demonstrate that they have had a positive impact on reducing the levels of GHG emissions, a "baseline scenario," which would depict the levels of emissions as they would be without the project.

All projects adopted by the NRCC are to comply with a variety of standards recognized globally, such as Verra (VCS) [3], Gold Standard [4], or the CDM [11]. This makes the credits valid, quantifiable, and accredited all over the world.

ii. **Key Carbon Accounting Concepts in the UAE Context**

To qualify for carbon credit issuance, renewable energy projects must adhere to several core principles:

- **Additional**ly, the project must demonstrate that it would not be financially or operationally viable without the revenue from carbon credits.
- **Baseline:** A reference scenario is developed to estimate emissions that would occur without the project (typically, the grid emissions factor).
- **Leakage:** The project must ensure that there are no indirect increases in emissions outside the project boundary due to its implementation.
- **Permanence:** Unlike forestry initiatives, renewable energy projects do not run the risk of reversing their carbon reductions; instead, sustained impact requires long-term operation and maintenance.

These principles are crucial, particularly in the UAE, where such sources may receive government backing and are executed in a competitive and liberalized power market. It also becomes complicated to prove additionality – the fact that it would not have occurred but for carbon credit revenue – where this is not backed by sufficient documentation.

iii. **Barriers and Enablers**

Although the role of the NRCC is promising, there are a number of obstacles, which might impede registration of renewable projects in the NRCC. These are high costs of verification, the lower availability of local expertise and regulator risks that pose a burden to developers willing to engage in the carbon credit business.

- High MRV costs for small-scale projects
- Limited local expertise in carbon credit documentation and validation

- Uncertainty in voluntary carbon prices may reduce financial predictability
- Awareness gaps among project developers and investors

However, several enablers are helping to address these challenges:

- The NRCC's digital platform simplifies registration, tracking, and credit issuance
- Government support and regulatory clarity from MOCCAE [1]
- Technical guidance and public-private partnerships to lower participation costs
- Growing private sector interest in offsetting and ESG-aligned investment

Case Study

This study evaluates the techno-economic feasibility and carbon reduction potential of adding 100 MW of utility-scale PV in the UAE. The efficiency of the system is ranked at 21%, and the system produces about 183,960 MWh of electricity per year. In the case of electric energy, the grid emission factor is 0.41t CO_2/MWh, and the emission of CO_2 is estimated to be reduced by 75,423.6 t/t/t/t/t/ t/annum. Therefore, taking into account a minimum carbon credit price of $ 10/tCO_2, the carbon revenue will be $ 754,236 per year. The primary source of revenue is the sale of electricity at $62 per MWh, which is based on the green band tariff of Dubai [2]. The CAPEX required will be $ 90 million, while the annual OPEX is $ 1.5 million, which will cost 1.5% of the CAPEX. At a project life of 25 years and a discount rate of 6%, carbon and energy revenue have been taken into consideration in order to estimate the project's profitability. The performance highlights the usefulness of carbon credit as an additional avenue of increasing the break-even of the renewable power infrastructure, thus propelling the UAE's vision towards energy sustainability and emissions cut [1].

Table 3—PV- Case Study Parameters

Parameters	Value	Notes
Installed Capacity (MW)	100	Utility-scale PV
Capacity factor	21%	Assumed
Annual Wh) energy output	183960	
Grid Emission Factor O_2/MWh)	0.41	As per IRENA/UAE reports
Carbon Credits Generated O_2/yr)	75423.6	
Carbon tCO_2) Credit Price	10	Conservative base case
Credit Revenue ($/yr)	754,236	Revenue from carbon credits
Project Lifespan (yr)	25	Typical PV project lifetime
CAPEX (M$)	90	Based on $900/kW
OPEX (M$/yr)	1.5	1.5% of CAPEX annually
Discount Rate	6%	Used for NPV & DPP calculation
Electricity price ($/MWh - bai price in green band)	62	-

Financial Impact Analysis

LCOE

Making utility-scale (PV) projects financially feasible is greatly impacted by carbon pricing, especially with the new climate policy instrument. As illustrated in Figure 1, the carbon prices and the LCOE exhibit a clear inverse relationship. LCOE of wind power declines with the increasing carbon price, since the economic value in the form of carbon credits increases proportionally. Such dynamics highlighted the need to incorporate carbon pricing mechanisms in renewable energy projects.

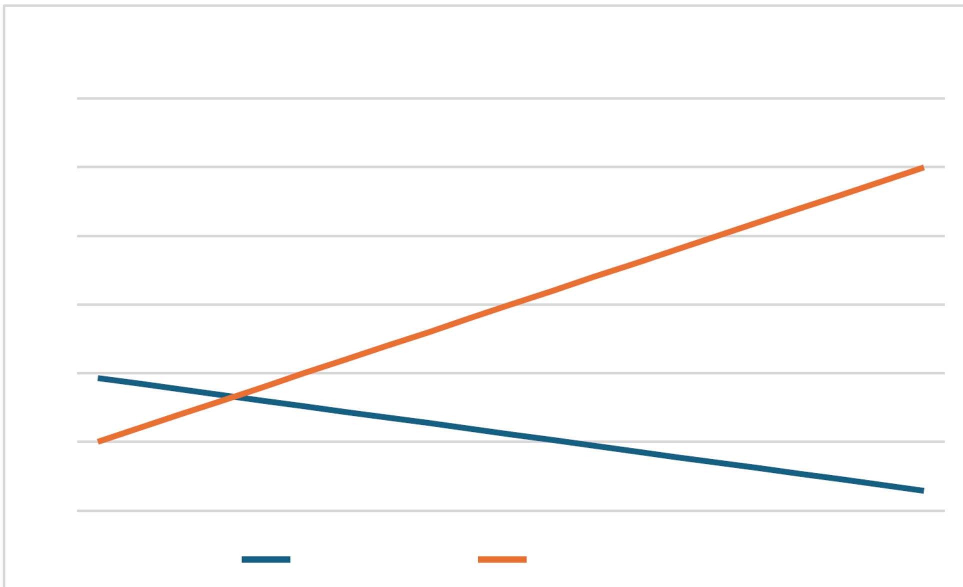

Figure 1—Carbon Tax vs. LCOE

Hypothesis

The hypothesis of the given research is that when the carbon revenue is generated due to the sales of carbon credits, it provides some extra income that may cover the entire consumption of the energy production costs. This additional revenue leads to a high LCOE, hence enhancing the feasibility of renewable energy projects in terms of finance. It is more pronounced when there is an increase in the prices of conventional energy sources as a result of rising carbon regulations. As an example, an upward adjustment of the carbon prices in greenhouse work, because of the increasing revenues on credits, leads to the prevention of operational expenses and a better income on investment (ROI) in the project.

Also, carbon pricing can be used as a tool to promote new renewable energy projects in the energy market by enabling them to be more competitive. Carbon credit revenue assists in stabilizing the project funding, the energy price volatility, and the dependency on the government subsidies or the fossil fuels-based power. The outcome will be a more competitive renewable energy market, more economically viable in the long-term, and a greater drive towards low-carbon energy systems [1][2][5].

In this graph, the negative correlation between the carbon prices and the LCOE has been shown. When carbon prices are increased the LCOE reduces and shows the financial implication on renewable energy projects by the carbon tax.

IRR

Figure 2 represents the correlation between the IRR and the change in the price of carbon. Based on the graph, the financing value of renewable energy is on an upward trend as the size of the carbon price increases. Therefore, the carbon price is essential in promoting the financial development of renewable energy investments.

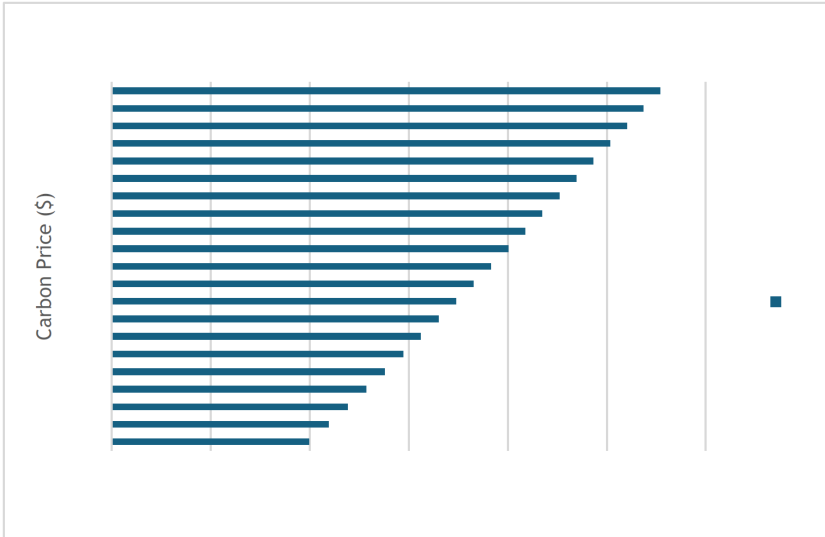

Figure 2—IRR vs. Carbon Price

The analysis shows a strong positive correlation between carbon price and IRR, and this points to the fact that there is a positive relationship between the carbon price and the profitability of the project. When the carbon price is listed at a fundamental rate of $10/t$CO_2$, the IRR is less substantial, indicating that the carbon revenues are insufficient to provide the project with a significant financial advantage. Nonetheless, the IRR is also very sensitive and rises at high carbon prices, especially when carbon prices increase beyond 100-200/tCO_2, which is the optimal carbon price range expected for the second half of this century. This indicates a significant increase in project profitability, highlighting the vulnerability of renewable energy projects to carbon pricing initiatives.

Hypothesis

In this research, one of the hypotheses is that there exists a high positive relationship between carbon price and (IRR) in renewable energy projects. The interdependence is fuelled by various sources of income that are earned through the sale of carbon credits. With an increment in the value of carbon, the financial advantages of renewable energy schemes increase as it decreases the LCOE and thereby increases the financial benefit as well as the economic viability of the project. The higher revenues in terms of carbon credit sales provide more opportunities to invest in the project and thus support the business case of renewable energy projects and streamline the process towards sustainable energy.

In addition, the increased level of carbon prices lowers stranded-asset risk, which is a predicament facing fossil fuel-based sectors mostly, thereby making renewable energy investments more appealing. This trend also concludes that carbon pricing is vital to enhance the development of a green economy, and encourage the long-term financial investment in low-carbon technologies [5][6].

Collectively, the results point out the opportunities that exist in using carbon pricing as a mechanism to lower risk in the investment of renewable energy sources. The integration of the carbon market incentives not only facilitates capital recovery but also ensures that the models align the financial performance with climate policy goals. This supports the fact that the UAE and similar economies should bolster their domestic carbon credit markets as a strategic means to spur the growth of low-carbon infrastructure.

Discounted payback

Table 4—Carbon price effect on discounted payback

Discounted Payback	**~14 years**	**~12.2 years**	**~9 years**	**~6 years**	**~5 years**
Carbon price ($/t$CO_2$)	**0**	10	60	100	200

(DPP) is very crucial in assessing the feasibility and the risk of the rising capital-intensive renewable energy projects is the (DPP). For this purpose, the outlined investigation was conducted on the DPP of a 100MW utility-scale (PV) installation under different carbon price levels to determine the impact of carbon revenue on investment payback periods.

In the absence of any carbon pricing mechanism (price of carbon = $0/t$CO_2$), the Project NPV and DPP of the project stand at roughly 14 years, which can avert private sector investments as per conventional micro finance tools and the discounted cash flow rate of returns. Therefore, as carbon pricing mechanisms are incorporated, much enhancement of the project bankability is reported. However, at the conservative price of carbon $10/t$CO_2$, the DPP decreases to 12.2 years, which suggests a moderate improvement in the company's financial condition. Further, the DPP shows a more drastic reduction at the mid-range of carbon price: with the carbon price tag of $60/t$CO_2$, the DPP stands at circa nine years; it further falls to approximately 6 years in case of $100/t$CO_2$.

Notably, under a high-carbon price scenario of $200/t$CO_2$ aligned with potential mid-century climate policy projections, the DPP drops to just 5 years. This sharp decline was mainly attributable to the fact that carbon pricing plays a critical role in the improvement of capital recovery, risk management, and the development of competitive renewable power projects.

These results highlighted the financial savvy in carbon markets that are available to deliver financing. Integrating carbon credit revenue into the financial plan helps boost payback conditions in terms of efficiency for solar PV projects, thus connecting environmental value with investor gain. For this reason, stable and predictable policy frameworks on carbon prices could transform the markets for renewable energy in the UAE and other emerging countries[1][2][5].

Sensitivity Analysis

Sensitivity of Financial Metrics to Discount Rate: A Carbon-Inclusive Perspective

Table 5—Discount rate sensitivity analysis

Discount rate	**6%**	**8%**	**10%**
Annual Net Cash Flow ($)	10659756	10659756	10659756
NPV (25 yrs)	46267457.51	23790509.52	6759031.797
LCOE ($/MWh)	42.32533507	49.88505172	57.95222058

This study examines the effect of the applicable discount rate on the profitability of utilizing a 100 MW (PV) project, taking into consideration the effects of carbon credits. To arrive at different savings rates in Situations 2 and 3 than in Situation 1, three discount rates of 6%, 8%, and 10% are used to represent investors' expectations and market conditions in the respective situations.

When the discount rate is 6%, the evaluated metrics reveal the highest improvement for the project. The NPV after 25 years is approximately equal to $46,27 million, and the DPP is again slashed to nearly 12,2 years. LCOE is also at the lowest at $42.33 MWh for the same reasons: low cost of capital and better estimation of the value of future cash inflows.

At an 8 % discount rate, the NPV will reduce to $23.79 million, the payback period to about $15 years, and the LCOE to $49.89 MW.h. However, if the discount rate is scaled up to 10%, then the project's NPV declines to $6.76 million, the payback period increases to about 20 years, and the LCOE is raised to $57.95/ MWh.

These results further show that the cost of renewable energy projects is sensitive to the discount rate. With the rise in the discount rate, the value of cash flows received in the distant future decreases, and the project's appeal declines. However, the integration of carbon credits as an additional revenue source reduces the impact of the above problems. It enhances the status of the annual net cash flow and the feasibility of the company's investment decisions in all the forecasts [1][2].

These results show that there is a need to create enabling policy conditions and financing mechanisms to make capital investment in clean power projects cheaper. In particular, investment profitability may be enhanced manifold by the implementation of stable carbon credits markets, as in the case operated by the NRCC. The financial risk associated with renewable energy projects can be mitigated by the NRCC, which provides a structure for creating, selling, and verifying carbon offsets. Further, NRCC could help establish a desirable policy framework, such as carbon price floors and far-reen purchase credit agreements, to stabilize revenue flows and encourage more private investment. Such initiatives will not only facilitate the movement towards a low-carbon energy sector in the UAE but also increase clean energy investments globally [5] [6].

Carbon credit effect on nuclear energy financials

This section aims to determine the impact of carbon credits on the profitability of nuclear power plants. This will be done by analyzing various aspects, such as CFR based on NPV, LCOE, and DPP. Given the extensive capital investment and the long years it takes for facilities like these to repay the invested cost, any possible carbon revenue streams thus have a high impact on improving overall project bankability.

The model applies the suppositions of a conventional large power generation structure with high fixed costs, low marginal costs, carbon emissions, and stable generation volumes in a large- scale nuclear power plant. Without carbon pricing, the project's NPV remains marginal or negative based on a 60-year operating duration and varies significantly for different discount rates. However, when carbon pricing is introduced—and its range is between \$10/tCO_2 and \$200/tCO_2— a steady enhancement of all the key financial parameters under consideration is also recorded [7][8].

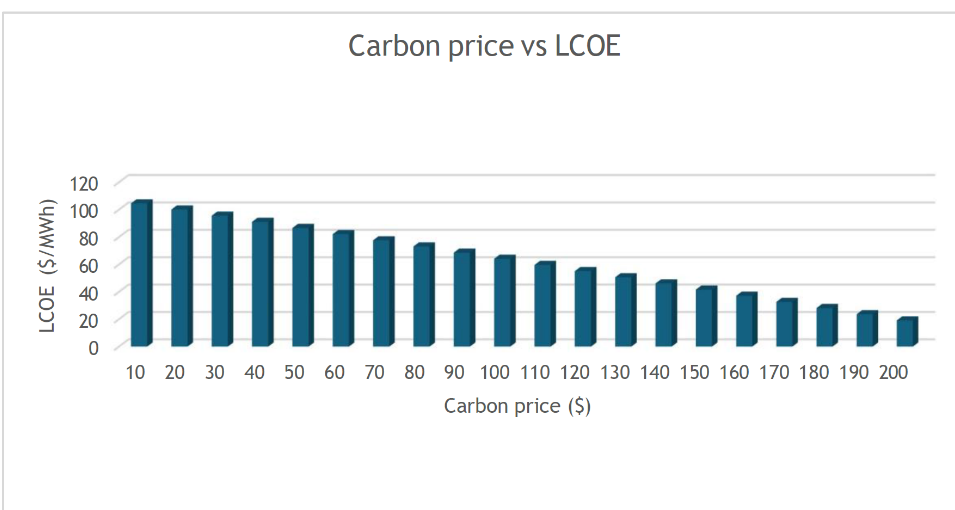

Figure 3—Carbon price vs LCOE

For example, at the \$30/tCO_2 carbon price, the latter adds value to electricity income, reduces the LCOE considerably, and brings the cost really closer to the market price. At \$100/tCO_2, the LCOE is even at par with conventional fossil fuel sources, and the DPP is greatly improved by over one and a half decades in reference to the baseline case. NPV under such a regime rises to significant levels; thus, the investment case is improved even for relatively higher discount rates.

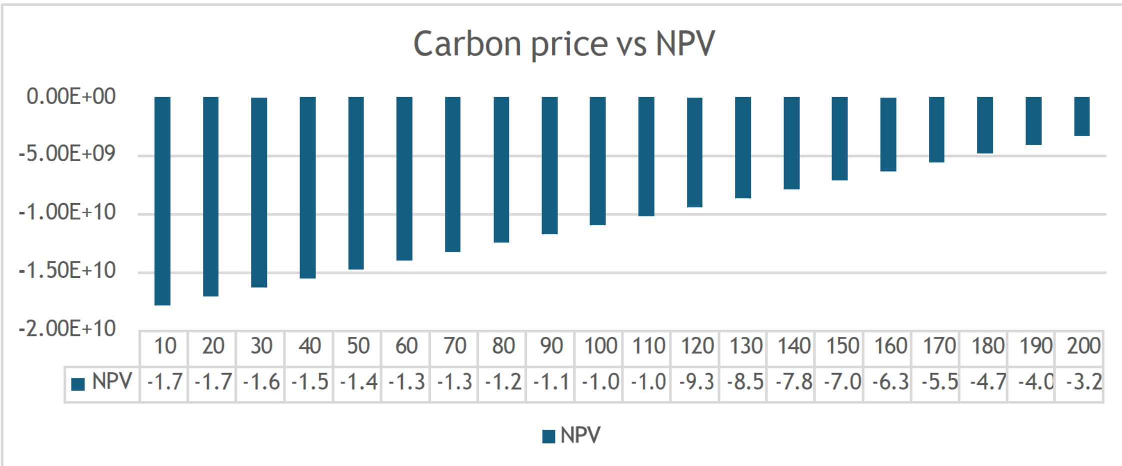

Figure 4—Carbon price vs NPV

These findings support the role of nuclear energy in the development of decarbonized power systems. Since it has mostly zero carbon in its operation and offers baseload electricity, nuclear power is likely to gain in function from higher carbon credits. The use of carbon credit mechanisms in the financing of nuclear projects provides a means of insurance on the investment, promotes a decrease in subsidy support dependency, and achieves both national energy security goals and climate change targets. Thus, establishing efficient carbon price policies also promotes not only renewable energy technologies but also good, dispatchable, and sustainable types of generation, like nuclear.

Comparative Financial Impact of Carbon Credits in the EU and UAE

Carbon credit systems have vastly different effects on the financial viability of green energy projects in the EU and the UAE because of the disparities in market maturity, price structures, and the set of rules binding in these two markets, and because the nature of the regulatory rationalizations that find utility in the EU and the UAE are not comparable. The EU Emissions Trading System (EU ETS) is the most advanced and live carbon market among European countries in the global market. It supplies prices that recover costs, based on rules that typically fall between 80 and 100 euros per ton of CO_2, around 85 and 105 dollars at the current exchange rate [9]. It is a very favorable pricing regime that ensures very high financial returns from renewable energy projects. On the contrary, the UAE's carbon pricing is lower due to the carbon market being in a developing phase, volatile, and lacking a fully viable compliance market. The addition of carbon credit revenues to financial models therefore causes the Net Present Value (NPV), the Internal Rate of Return (IRR), and Levelised Cost of Energy (LCOE) to improve, and the LCOE to drop by up to 30-40 %. Additionally, integrating carbon credits enhances project bankability by enabling discounted payback periods of 5-10 years across various project types.

The practice of carbon trading in the United Arab Emirates is still in its nascent stage compared to Australia. Starting with the registration of the National Carbon Registry, the UAE has begun the framework of the voluntary carbon trading system. However, as of now, the current price of carbon is still low, fluctuating between five and twenty US dollars per tonne depending on the project standard or for a specific buyer [6]. Specifically, there is low liquidity, which, together with relatively high price volatility in the market, means that carbon revenues cannot be integrated into the company's financial outlook with high reliability. Thus, the incorporation of carbon credits in the context of the UAE results in minor improvements in project viability, including a 2–5% rise in the IRR and a 10–20% reduction in LCOE and NPV values; nonetheless, the overall effect is less apparent than in the EU context. Years of discounted payback may be slashed by 1 to 3 years, but the lack of a compliance market provides investors with short-term faith and reasonable revenue predictability.

Challenges and Policy Gaps

Verification and Transaction Costs

One of the key issues that the UAE has to overcome in its carbon credit market includes high verification and transaction costs. MRV is important for providing credibility to the reports for achieving the new ambitious energy targets. Still, they come with corresponding costs in terms of finance and administrative efforts, particularly for small and medium renewable energy developers. These costs consist of employing the services of third-party verifiers as well as meeting the requirements of such international standards as Verra [3] or Gold Standard [4], among others, and handling issuance and registry costs. Since the carbon market is still developing in the UAE, activities such as the absence of standardized digital MRV tools and limited accreditation within the region increase these expenses. Thus, such project developers are unable to participate, meaning that this market is not experiencing faster growth.

Lastly, according to the results gathered in the case study above, the MRV (Monitoring, Reporting, and Verification) cost has generally ranged as shown in the table below for the renewable energy projects. These rates vary significantly by project size, and the verification cost was found to be disproportionately higher for small developers compared to larger projects. This is significant when viewed in the financial viability of the renewable energy projects, particularly among smaller developers, because the costs take a higher percentage of the total project costs.

Table 6—Verification cost

Project Type	Average Verification Cost (USD)	% of Total Project Cost
Small Solar PV (<1 MW)	$15,000 - $25,000	10–20%
Medium RE (1–10 MW)	$25,000 - $40,000	5–10%
Large RE (>50 MW)	$50,000+	<2%

Source: Aggregated estimates from Verra, UNFCCC CDM Toolkits, and IETA Market Readiness Reports.

Carbon Credit Price Volatility and Demand Uncertainty

The significant challenges in carbon markets, especially within the voluntary systems that the UAE has recently established through the registry, are fluctuations in prices and the constantly changing demand for carbon credits. There are a few drivers that can affect the price, such as fluctuations in the global market, new regulations, or when buyers are committed to sustainability goals. This is a significant issue since carbon credits are unpredictable and cannot be relied upon for financial planning, unlike earnings from project development. The absence of any long-term carbon price or guaranteed purchase in the form of such contracts, such as the forward value, leaves little basis for a sound business case for carbon-cutting technologies. For the firms operating in the UAE, where the local demand for carbon credits is still in the process of formation, this risk is even more significant, which may relieve the renewable energy developers from joining the program in the beginning [6][7].

Institutional and Developer Capacity Constraints

The effectiveness of a carbon credit system is not only defined by the policy framework but also relies more on the ability of government bodies and prospective developers. There are qualified personnel and companies, especially within emerging markets such as the UAE, with practical knowledge in the development of carbon projects, MRV processes, and global certification schemes. This has the effect of causing delays in project implementation, a lack of proper record-keeping, and calling for expensive experts from overseas.

On the institutional side, for instance, local expertise, although available, is inadequate across ministries and registries, and there is equally inadequate staffing, which makes it even challenging to streamline project approvals and ensure adequate operation of carbon registries. The legal and accounting preparedness, however, leads the list of the eleven factors constraining technical capacity. This involves the necessity of clear legal frameworks that regulate transactions involving carbon credits, ownership rights, and taxes, as well as the process for achieving this. Additionally, the accounting for carbon credits and emissions reductions must be clearly defined to ensure an accurate reporting method, effective regulation of emission reductions, and ease of integration into financial models. Such aspects are imperative to give developers and investors confidence in the carbon offset projects in terms of economic feasibility and legal acceptance.

Based on the gaps mentioned above, the following actions are suggested:

- Support local capacity-building programs that can help organizations improve their carbon accounting and MRV techniques.
- Encourage the UAE institutions to affiliate with the international carbon standard bodies and achieve partnership cooperation.
- National accreditation should be developed to prepare and qualify the local validators and verifiers.
- Develop clear legal and accounting frameworks to support cross-border carbon transactions and ensure alignment with Article 6.

Policy Recommendations for Enhancing the UAE's Carbon Credit Market

Considering the situations with the (NRCC) in the UAE and the lessons learned from the financial case study and cross-country comparisons, the following policy recommendations can be offered. This is because the following recommendations aim to address existing market challenges and improve the scalability and effectiveness of the NRCC in developing renewable energy projects in the UAE. Not only are these recommendations best practices, but they are also influenced by the unusual ideas from the case study of 100 MW (PV) projects and my belief regarding the additional enhancement of the carbon credit market in the country.

Introduce Standardized Baselines and Methodologies

The UAE should develop robust carbon offset methodologies at the highest level in non- conflicting sector areas, such as solar electricity, industrial energy efficiency, and waste-to-energy projects. These ought to adhere to global best practices, such as the Verra [3] and the Gold Standard [4], so that carbon credits produced in the UAE will be both credible and internationally recognized. Additionally, a special technical committee within the NRCC should be established to oversee the development, evaluation, and ongoing updates of these methodologies. The case study demonstrated the necessity of a well-organized baseline to compute emission reductions effectively, and this aspect should be addressed to promote market growth.

Establish Carbon Floor Prices and Long-Term Credit Purchase Agreements

The establishment of a carbon floor price by the government will set the price at a low point (e.g., USD $5-10/tCO_2e) and ensure that developers have a secure financial basis, mainly in the initial phases of carbon market development. Also, renewable energy projects can be supported by the introduction of long-term

(PPAs) that would encourage developers and generate stable revenues. By guaranteeing stable pricing mechanisms, the government will eliminate the uncertainty associated with carbon credit revenues, thereby making renewable energy projects more bankable and appealing to investors. This aligns with the financial analysis of the case study, which shows that even medium-priced carbon resulted in a significant increase in the IRR and project viability [1].

Scale MRV Automation through Digital MRV (dMRV) Platforms

The use of MRV platforms on a large scale in the UAE considerably reduces (MRV) expenses. The incorporation of the most futuristic technologies, including IoT and blockchain, would help the UAE improve transparency, real-time reporting, and data accuracy, thereby strengthening the reliability of carbon credits. The use of dMRV would have to be enforced in massive-scale projects, whereas smaller developers would need to be offered compensation or tax benefits to adopt dMRV usage. Such a policy would address the cost obstacles to MRV, which disproportionately affect small-scale developers, as identified in the case study [6], and enable the scalable growth of the carbon credit market.

Offer Financial Incentives to Encourage NRCC Participation

To attract more people to join the NRCC operation, the UAE government can offer special economic incentives by providing tax benefits to corporations (such as tax break deductions), waiving registration charges for industrial projects, or providing grants for projects demonstrating high environmental integrity. First-time developers or high-paced domains, such as waste-to- energy and industrial transitions, could also be assigned special incentives. Such incentives would reduce part of the initially incurred investments to participate and bring in more developers into the business of carbon credits. This plan will focus on the necessity of lowering the entry barrier and enhancing the financial appeal of participation in the NRCC, as discussed in the case study [5].

To sum it up, with these four focused measures, the UAE will be able to accelerate the development and growth of its carbon credit market, thereby advancing its progress toward a renewable energy transition and fulfilling its promise of climate action on a global scale. The approach presented, based on the case study and literature research, is feasible and necessary for establishing a scalable and effective carbon market in the UAE.

Conclusion

The NRCC of the UAE brings a new milestone in the country's carbon market and will introduce a potentially scalable and internationally accepted carbon trade solution. The case study of the 100 MW solar PV project reveals the potential of the NRCC to contribute immensely to the monetary feasibility of the renewable energy projects. Particularly, a moderate or higher carbon price (60-200/tCO_2) was found to enhance other key economic variables, including Internal Rate of Return (IRR) by 8-12%, Levelized Cost of Energy (LCOE) by up to 30%, and discounted payback period by more than five years, through carbon credit revenues, especially in moderate to higher carbon prices (60-200/tCO_2), the highest level of addition to their economy overall associated with carbon credit revenues [5] [6].

Nonetheless, numerous issues have impeded the performance of the NRCC. Significant expenses to conduct Monitoring and Reporting and Verification (MRV), especially on small-scale projects, and volatility of prices in the voluntary carbon market introduce financial risk to investors [6]. Additionally, the lack of a local trading capability and regulatory complications associated with geographical cross-trading hinder the growth of the carbon market during its development phase [5].

The UAE must enact specific changes to maximize the value of the NRCC. A carbon price floor (e.g., pledged USD 5-10/tCO_2) will give developers the needed certainty, and long-term renewable energy purchase agreements will encourage fresh investments. The use of digital MRV (dMRV) platforms would allow for compliance costs to be kept at reasonable levels and enhance data transparency, making the UAE compliant with international norms[7]. In addition, through this, the baselines of key sectors like solar energy and waste-to-energy will be standardized to increase market transparency and trust [3] [4].

Through such reforms, the UAE will stabilize its carbon market, ensuring investors are not at risk, and increase the sought-after projects in the renewable energy sector, which will ultimately result in investment spending by the private sector. The case study results demonstrate that the inclusion of carbon credits in renewable energy projects increases their financial performance, making them more bankable and in line with the country's long-term energy transition scheme [6].

The UAE has the advantage of being a leader and an investor in the carbon markets and renewable energy, particularly in the region. The UAE will achieve its Net-Zero 2050 goal and open a door to sustainable global energy through a thoughtful policy, a supportive institutional framework around the NRCC, and effective interventions in the market [1]— as an example, consider the case of Norway, which has been ranked the most globally competitive country in renewable energy related industries in 2019, with its current installed base of wind power energy amounting to 4,670 megawatts in 2022, to be followed by the establishment of the world-first-ever offshore wind power

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